

Task 2 Write-Up: Client Profitability & Spread Recommendation

1. Client Classification and Quantitative Justification

We classify clients based on their Expected Aggregate PnL (Agg PnL). If the Agg PnL is ≥ 0 , the client is **Profitable**. If it is < 0 , the client is **Costly**.

Profitable Clients:

- **Client A (Agg PnL = 1.897):** Highly profitable. Our PnL strictly increases over time, growing from 1.26 at $\tau = 5$ to 2.48 by $\tau = 30$.
- **Client B (Agg PnL = 1.631):** Very profitable. Shows steady PnL growth across all time horizons without dropping.
- **Client C (Agg PnL = 1.173):** Profitable and stable. The PnL stays consistently flat between 1.07 and 1.22 across all horizons.
- **Client D (Agg PnL = 0.299):** Marginally profitable. It is important to note that while the *average* is positive, the individual PnL actually drops into negative territory at $\tau = 25$ (-0.076) and $\tau = 30$ (-0.346). We barely break even overall.

Costly Clients:

- **Client E (Agg PnL = -0.401):** Costly. Our trades start profitable at $\tau = 5$, but drop below zero by $\tau = 15$ (-0.108) and fall steadily to -1.61 by $\tau = 30$.
- **Client F (Agg PnL = -1.440):** Highly costly (Toxic flow). This client forces us into a net loss almost immediately ($\tau = 10$ PnL is -0.176) and causes massive losses by $\tau = 30$ (-3.629).

2. Relationship Between δ^* and the Adversity Profile

The minimum half-spread (δ^*) acts as an **insurance premium**. It is the exact price buffer we must add to our quotes to protect ourselves against adverse price movements.

There is a direct, positive correlation between a client's Adversity Profile (Task 1) and their required δ^* (Task 2):

- **Low Adversity = Zero Premium:** Clients A, B, and C had low adversity profiles (well below 50%). Because the market rarely moves against us when trading with them, our standard spread is already enough to make a profit. Therefore, their required δ^* is **0.000**.
- **Moderate Adversity = Small Premium:** Client D crossed the 50% adversity threshold late in the time horizon. Because they cause slight adverse market movements, we must quote them a slightly wider spread ($\delta^* = 0.0096$) just to break even.
- **High Adversity = Massive Premium:** Clients E and F had very toxic adversity profiles (rising steeply to 54.8% and 61.8%). Because we know the market price will aggressively move against us after they trade, we must charge them a much larger upfront fee to offset the guaranteed future loss. Client E requires a δ^* of **0.0189**, and Client F requires the highest δ^* of **0.0328**.